Construction and Building Materials 514 (2026) 145564

Contents lists available at ScienceDirect

Construction and Building Materials

journal homepage: www.elsevier.com/locate/conbuildmat

# Damage evaluation of regolith-resin-composite (RRC) materials for lunar structures using linear and nonlinear ultrasonic waves

Md Maruf Molla, Ching Tai Ng *, Tafsir Tafsirojjaman, Scott T. Smith

*Andy Thomas Centre for Space Resources, School of Civil Engineering and Construction Management, Adelaide University, South Australia 5005, Australia*

ARTICLE INFO

*Keywords:*
Non-destructive evaluation
Regolith-resin-composite
Compressive loading
Ultrasonic wave
Second harmonic generation

ABSTRACT

The growing interest in establishing permanent bases on the moon necessitates the development of durable construction materials that can withstand the extreme lunar environment. Even minor damage from repeated loading, material aging, and harsh conditions can disrupt the controlled internal environments that lunar structures create and thus pose serious threats to potential human inhabitants. Non-destructive inspection is essential for ensuring a safe and secure habitat. Although the ultrasonic wave technique is a commonly used non-destructive inspection method, the behaviour of ultrasonic waves in extraterrestrial construction materials has not been fully explored. This paper investigates the viability of ultrasonic waves and their performance for non-destructive evaluation in a novel material fabricated using lunar regolith simulant and epoxy resin, known as regolith-resin-composite (RRC). An experimental framework is employed that comprises nine loading stages, each of which involves applying a load at a specific percentage of the material's ultimate capacity, followed by unloading. Subsequently, the linear and nonlinear characteristics of the ultrasonic wave are investigated at the end of each stage. Linear ultrasonic parameters, such as wave velocity and amplitude and a relative nonlinear ultrasonic parameter are then derived by analysing the received signal. All ultrasonic parameters are presented in relation to increasing compression load on RRC samples. The results indicate that the relative nonlinear parameter increases by 76 % at the maximum applied load. In comparison, the maximum change of the linear parameter (amplitude) corresponding to that load is 19.9 %. Thus, the second harmonic generation is highly sensitive to damage progression and exhibits distinct values at different loading stages in the RRC, thereby signifying the reliability of ultrasonic waves for inspecting lunar construction materials.

## 1. Introduction

### 1.1. Lunar construction

The establishment of a permanent presence on extraterrestrial bodies, with the Moon expected to be the first site for crewed settlements, is gaining significant momentum. Successful human presence on the Moon requires robust infrastructure, which demands innovative materials capable of withstanding extreme lunar environmental conditions. The harsh environment demands materials with superior properties, such as high density, strength, freeze-thaw resistance, and radiation protection – characteristics that differ significantly from those required for Earth-based construction [1–3]. Additionally, the cost of transporting materials from Earth to the Moon highlights the need to utilise in-situ lunar materials for construction [4–6]. The abundance of lunar regolith, its ease of extraction, resistance to the Moon's freeze-thaw cycles, and ability to shield against radiation make it a promising material for lunar construction [7–10]. To investigate the practical aspects of lunar habitation, regolith simulants with mineralogies similar to those of the lunar regolith collected during several Apollo missions have been developed to date [1,11].

In the literature, the construction materials studied that are based on regolith simulant include sulphur concrete [12], sintered and melted regolith particles [13], Portland cement blended mixture [14], resin-based polymer-bound regolith [15], and geopolymer-based concrete [16]. Owing to the high energy demand of sintering, the low melting point of sulphur concrete, the presence of significant porosity in Portland cement–based mixtures, and the water requirement of geopolymer concrete, waterless or nearly waterless regolith-resin-composite (RRC) emerges as a viable candidate [15,17,

* Corresponding author.
*E-mail addresses:* mdmaruf.molla@adelaide.edu.au (M.M. Molla), alex.ng@adelaide.edu.au (C.T. Ng), tafsirojjaman@adelaide.edu.au (T. Tafsirojjaman), scott.smith@adelaide.edu.au (S.T. Smith).

https://doi.org/10.1016/j.conbuildmat.2026.145564
Received 21 October 2025; Received in revised form 24 January 2026; Accepted 8 February 2026
Available online 14 February 2026

18]. Over the past decade, numerous studies have been conducted to develop RRC materials for use in lunar construction. Lee et al. successfully developed polymeric concrete in 2015 using 10 % thermoplastic polymer and 90 % lunar simulant. The process involved high-temperature (230℃) heating to harden the mixture, resulting in a compressive strength of 12.9 MPa [19]. Afterwards, Caluk et al. [15] and Tafsirojjaman et al. [20] utilised high resin content, resulting in superior mechanical and microstructural properties. In contrast, Chen et al. [21] and Oh et al. [22] developed RRC with lower polymer content but employed high compaction pressures (up to 350 MPa) to densify the mixes during casting. To address this, Hossain et al. [23] proposed a detailed investigation focused on minimising both compaction effort and resin content. They reported that RRC with 10–15 % resin content produced a closely packed microstructure with consistent resin-regolith interaction, leading to lower porosity. The material showed no weight loss at 180℃ in thermal analysis. Hence, such compacted RRC material has been considered in the current study.

The utmost priority in lunar construction is ensuring a safe and secure environment for human inhabitants inside the infrastructure. From this perspective, non-destructive inspection of the lunar structure is vital, as even minor defects could pose a significant risk to lives. Therefore, there is a strong motivation to develop and effectively incorporate damage inspection techniques to support lunar construction and ensure there is no failure or leakage of the structures.

### 1.2. Linear and nonlinear ultrasonic techniques

In the last two decades, ultrasonic techniques have been developed for damage inspection, enabling the detection of surface, subsurface, and internal damage in materials at both microscopic and macroscopic levels [24–26]. This method can detect a wide range of damage, such as delamination, debonding, corrosion, fatigue, and thermal degradation [27–31]. These techniques have been applied to a range of materials, such as metals and non-metals [32,33]. A recent review by Xu et al. demonstrated the effectiveness of ultrasonic wave-based techniques for evaluating the conditions of construction materials. The review highlighted ultrasonic methods for identifying defects in concrete and other building materials and presented an improved algorithm with flexible probe configurations [33]. Ultrasonic waves have been utilised not only to detect but also to locate and quantify the extent of damage. This is achieved by analysing time-domain, frequency-domain, or time-frequency data, which provide key damage features for damage detection [34]. Ultrasonic characteristics have also been widely used to assess structural safety and evaluate material properties. These include linear features such as time-of-flight, wave velocity, and amplitude variations obtained from the received time-domain signal. Another ultrasonic technique is Resonance Ultrasound Spectroscopy (RUS), which is based on resonant frequencies of materials and can be correlated with their dynamic elastic modulus [35,36]. However, this technique is mainly utilised for structures composed of homogeneous materials and featuring simple geometries. The linear features of ultrasonic waves reflected from damage in the material may remain unchanged during the initial stages of damage progression, as they are not sensitive to micro-level defects. This limitation exists because their detection efficiency is constrained by the wavelength of the corresponding wave [37]. Hence, to effectively detect early-stage damage, especially those micro-level defects, more advanced features are required.

Unlike linear features, nonlinear features of ultrasonic waves can provide valuable insights into early-stage detection of micro-level defects in materials. The technique extracts nonlinear responses resulting from the intrinsic or damage-induced characteristics of the materials. Several mechanisms can lead to these nonlinear behaviours, which are usually classified into two categories. The first is material nonlinearity, commonly associated with the early degradation of bulk material behaviour, stemming from factors such as crystal dislocations, microvoids, and initial plastic deformation. The second is contact acoustic nonlinearity (CAN), which relates to dynamic interactions at material interfaces, including delamination and debonding, showcasing nonclassical characteristics such as clapping and frictional slip. Hence, these effects lead to various ultrasonic phenomena, including signal modulation, harmonics, waveform distortion, and changes in velocity [38]. In the literature, researchers have proposed various nonlinear features for damage detection by studying different ultrasonic phenomena. Wave mixing [39], sideband generation [40] and higher-harmonic generation [41] are the most commonly utilised methods for identifying damage in structures. Among these nonlinear features, the generation of higher harmonics has been extensively investigated for damage detection in various materials. Higher harmonics are generated due to the distortion of elastic waves resulting from their interaction with nonlinear elastic materials. These higher harmonics may include the second harmonic and, in certain cases, extend to the third harmonic. Notably, the second harmonic is particularly effective for early microscopic damage detection in materials, demonstrating its effectiveness across various loading and unloading scenarios [42–45]. Researchers have explored the application of harmonic generation for damage evaluation in materials. Ongpeng et al. investigated higher harmonic generation in concrete by applying various load patterns at different water-to-cement ratios [46]. Kim et al. utilised second harmonic generation of ultrasonic waves to evaluate time-dependent drying shrinkage in concrete [47]. Zhao et al. conducted extensive studies on second- and third-harmonic generation to evaluate early-stage damage in concrete. Their research also detected semi-penetrating cracks with different angles on concrete surfaces. The study concluded that the harmonic generation method is particularly suitable for long-term structural monitoring, as it eliminates the necessity for frequent changes in the detection position [48]. Additionally, Chen et al. studied the second harmonic generation to differentiate between various stress levels in concrete post-tensioning strands, considering the changes in length. A correlation was established between the nonlinearity parameter derived from the second harmonic generation and the tensile stress in the strand [49]. Overall, second harmonic generation has emerged as a promising method for detecting early-stage damage in materials [37,50–55]. This is particularly important for lunar structures, where even the smallest damage or imperfection can lead to catastrophic failure due to the harsh lunar environment (e.g., vacuum, radiation, extreme thermal cycles).

To date, the literature has not fully investigated damage detection of lunar construction materials using ultrasonic wave phenomena. Second harmonic generation may be a viable option for inspecting lunar regolith-based materials. While the technique is effective for damage detection in conventional Earth-bound concrete, its applicability in lunar construction necessitates a thorough investigation before it can be considered reliable. This is because the chemical composition of Earth-based construction materials differs significantly from that of lunar regolith-based materials [9,15]. Conventional concrete and resin-based composites differ significantly in their binding media, which greatly influence their mechanical and dynamic responses. The bonding material in resin-based composites is epoxy, which is considerably stronger than the binders used in conventional concrete. Cement-bound concrete often contains coarse and fine aggregates, which form multiple interfacial surfaces with the binder, resulting in a brittle matrix with relatively low intrinsic viscoelasticity. Therefore, energy dissipation is primarily dominated by friction at microcracks and slip at the aggregate-matrix interface in conventional concrete. The polymeric binder surrounding the aggregate particle in resin-based composites makes the interface more viscoelastic than traditional concrete [56]. Additionally, the regolith used to prepare lunar construction materials consists of fine particles, with over 90 % of them smaller than 0.3 mm [57]. These fundamental material differences may lead to distinct micro- and macro-behaviour under loading, thereby affecting wave propagation, scattering, and damage identification through the material. The higher viscoelastic nature of the particles in the polymer composite makes wave propagation more challenging due to increased absorption of vibrational energy. However, it is a desirable property that significantly enhances resistance to high-velocity micrometeorite impacts [58]. Therefore, the ultrasonic wave phenomenon as it propagates through the RRC material should be further investigated. Moreover, RRC made with epoxy resin exhibits a considerably lower longitudinal wave velocity due to its lower stiffness than traditional construction materials [15]. This results in a shorter wavelength, which can benefit linear ultrasonic applications. Therefore, both linear and nonlinear ultrasonic characteristics should be examined in lunar regolith-based construction materials.

The primary objective of this study is to investigate the phenomenon of ultrasonic bulk waves propagating through the newly developed RRC material. Furthermore, an experimental investigation is conducted to evaluate different damage levels caused by compressive loading in this material, utilising both the linear and nonlinear features of ultrasonic bulk waves. Additionally, the study evaluates the sensitivity and reliability of ultrasonic parameters under varying compressive loads applied to the materials.

## 2. Second harmonic generation of ultrasonic waves

There are two different types of ultrasonic approaches, as mentioned in the introduction, linear and nonlinear. Linear ultrasonics is based on features such as wave velocity and amplitude, where the wave response is proportional to the input energy. In contrast, nonlinear ultrasonics uses the inherent material nonlinearity to detect microdamage through features like harmonic generation. The second harmonic corresponds to the double frequency. This paper is mainly focused on the second harmonic of the nonlinear features.

It is natural for materials to contain pre-existing microcracks, which result in a nonlinear stress ($\sigma$) – strain ($\epsilon$) relationship. The nonlinearity in the stress-strain relationship varies with changes in the material's microstructure, such as degradation or damage progression. When an elastic wave propagates through a nonlinear material, it may become distorted and take on various forms. A well-known mechanism is the redistribution of wave energy to higher frequencies, which can produce higher harmonics, although the excitation is usually limited to a single frequency [30,53,59,60]. In most cases, the magnitude of the generated third or higher-order harmonic is much smaller. This study, therefore, focuses on the second harmonic generation [38]. To understand the second harmonic generation, we consider a single-frequency longitudinal wave that propagates in a nonlinear solid medium. In classical linear elasticity, the stress ($\sigma$) and strain ($\varepsilon$) are related proportionally according to Hook's law. However, in nonlinear elasticity, the stress and strain relationship can be expressed by the nonlinear form of Hook's law as shown in Eq. (1) [61,62].

$$\sigma = E_0\varepsilon(1+\beta\varepsilon+\gamma\epsilon^2+\dots) \tag{1}$$

where $E_0$ is the material linear elastic modulus, and $\beta$ and $\gamma$ are second and third-order nonlinear elastic constants. This study considers terms up to the second order in the series, with $\beta$ being referred to as the nonlinearity parameter in the following sections. Assuming negligible attenuation, the one-dimensional equation for the propagation of a longitudinal wave through the solid medium can be expressed as:

$$\rho_0\frac{\partial^2 u}{\partial t^2} = \frac{\partial\sigma}{\partial x} \tag{2}$$

where $\rho_0$ is the material density, $u$ is the particle displacement at time duration t, and $x$ is the wave propagation distance. The following governing equation of the propagating nonlinear wave can be derived by substituting $\sigma$ from Eq. (1) into Eq. (2), considering terms up to the second order.

$$\rho_0\frac{\partial^2 u}{\partial t^2} = E_0\frac{\partial^2 u}{\partial x^2} + 2E_0\beta\frac{\partial u}{\partial x}\frac{\partial^2 u}{\partial x^2} \tag{3}$$

To obtain the final solution from Eq. (3), perturbation theory can be applied. Assuming the initial excitation signal is in the form of a sinusoidal wave with a single frequency [38].

$$u = \quad A_0\cos(kx-\omega t) - \frac{1}{8}\left(A_0^2k^2\beta x\right)\sin 2(kx-\omega t) \tag{4}$$

where $k$ is the wavenumber $\left(=\frac{\omega}{V_L}\right)$, in which $\omega$ is the angular frequency, and $V_L$ is the longitudinal wave velocity. The second term in Eq. (4) represents the second harmonic component. If $A_0$ is the amplitude corresponding to the fundamental component and $A_2$ is the amplitude corresponding to the second harmonic component, then

$$A_2 = \frac{1}{8}\left(A_0^2k^2\beta x\right) \tag{5}$$

$$\beta = \quad \frac{8}{k^2x}\frac{A_2}{A_0^2} \tag{6}$$

Since the frequency and propagation distance of the wave remain constant during the ultrasonic wave study after each conditioning of the material, the term $8/k^2x$ in Eq. (6) can be considered as constant. Therefore, the relative nonlinearity parameter $\beta'$ can be introduced here, which has a linear trend with respect to the absolute nonlinearity parameter, $\beta$.

$$\beta' = \frac{A_2}{A_0^2} \quad \propto\beta \tag{7}$$

Therefore, in this study, the relative nonlinearity parameter is used to analyse the growth of the second harmonic with increasing damage in RRC. An increase in nonlinearity due to second harmonics reflects the growth of micro-damage within the material. When ultrasonic waves interact with microcracks, the compressive portion of the wave tends to close the cracks and allows the wave to pass through, whereas the tensile portion opens the cracks and restricts wave transmission. This phenomenon is commonly referred to as the clapping effect. As a result, the effective wavelength is halved, and the frequency is doubled, leading to the generation of second harmonics. Consequently, higher microcrack density and severity increase wave-crack interactions, thereby increasing the relative nonlinearity parameter [37,38,50,61].

## 3. Methodology

A schematic representation of the methodology employed to investigate the ultrasonic parameters with respect to the varying levels of applied compressive load in this study is shown in Fig. 1. The linear parameters can be easily extracted from the received signal. To determine the nonlinear ultrasonic parameter, a single-frequency excitation ($f_0$) is used as the incident wave. When this wave interacts with a nonlinear material, it generates a second harmonic at frequency $2f_0$ in addition to the fundamental harmonic at frequency $f_0$. The relative nonlinearity parameter can also be validated by varying the input power level. As the input power increases both $A_0$ and $A_2$ exhibit a corresponding rise. By carefully adjusting the input power and measuring the resulting amplitude changes, a linear relationship can be established between $A_2$ and $A_0^2$. The slope of the linear fit represents the relative nonlinearity parameter. While material nonlinearity contributes to the results, system-related nonlinearity (such as from the amplifier, transmitter, and coupling media) also plays a role. However, system nonlinearity remains constant, while only material nonlinearity changes as damage progresses. Therefore, normalising the ultrasonic parameter relative to its value in the intact condition can effectively represent its response across different load levels. For consistent comparison, the

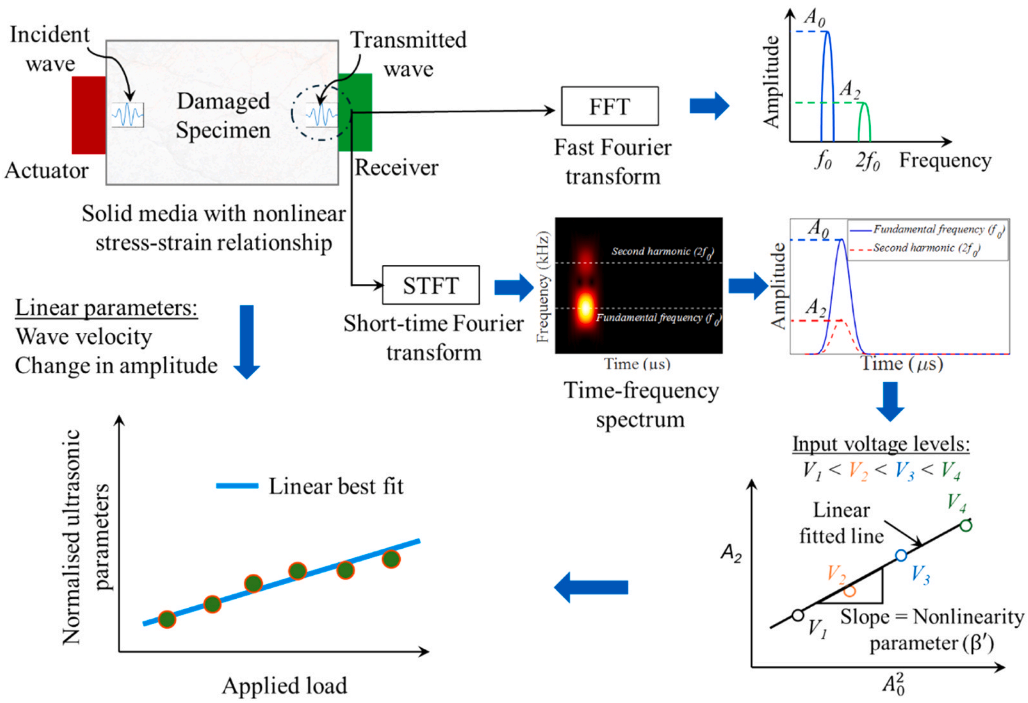



**Fig. 1.** Schematic diagram of the methodology for damage assessment using ultrasonic features.

frequency and wave propagation distance must remain the same throughout each measurement [24,44].

## 4. Experimental investigation and signal processing

### 4.1. Specimen preparation

The materials used in this study include lunar highlands simulant (LHS-1E), a high-fidelity simulant that closely replicates the mineralogical and physical properties of actual lunar soil [20], and a high-temperature thermosetting epoxy resin system (R2600 resin and H2409 hardener), which served as the binding matrix for fabricating the composite. Fig. 2(a) and (b) show the loose regolith simulant and epoxy resin, respectively.

The resin and hardener were mixed at a mass ratio of 100:35. Hossain et al. reported a maximum epoxy content of 15 % under a compaction pressure of 20 MPa, without any indication of epoxy bleeding, which yielded the highest strength [23]. Accordingly, to produce a homogeneous composite, 15 % epoxy resin by weight was blended with the regolith simulant using an electric mixing machine for three minutes, as illustrated in Fig. 2(c). The resulting mixture was then poured into rectangular moulds to prepare specimens. Due to the higher cost of materials (i.e., especially the regolith simulant), the specimen size was limited to $40 \times 40 \times 160$ mm$^3$ . After casting, the mixture was compacted under a compressive pressure of 20 MPa using a Universal Testing Machine (UTM). The specimens were left to cure at room temperature for 24 h before demoulding. Post-curing was then carried out in a laboratory furnace following a controlled thermal profile: 8 h at 50 °C, 3 h at 120 °C, 2 h at 150 °C, 1 h at 200 °C, and finally 1 h at 220 °C. The cured specimens were subsequently cut using a saw, and the prepared specimens are shown in Fig. 2(d). Three cube samples were used to determine compressive strength following ASTM C109 standard test method [63], while the remaining specimens were reserved for non-destructive evaluation. The compressive strength test, conducted under a displacement-controlled constant loading rate of 0.5 mm/min using the UTM, yielded an average compressive strength of 134.3 MPa.

### 4.2. Experimental setup and procedure for compressive loading

The comprehensive experimental program involved applying compressive loading to induce damage in the specimens, followed by an investigation of the interaction between ultrasonic bulk waves and the resulting damage. To ensure a systematic approach, the experimental procedures were organised chronologically, and they are discussed in detail in the following sub-sections.

A uniaxial compression load was applied to the specimen to induce damage using a UTM with a capacity of 1000 kN. A spherical pivoting platen, or swivel platen, consists of a hardened steel bearing that permits

(a) (b) (c) (d)

**Fig. 2.** Specimen preparation, (a) loose regolith simulant (LHS-1E), (b) epoxy resin, (c) electrical mixing machine, (d) specimens.

slight tilting to align with the specimen surface. This platen was positioned between the specimen and the load-transferring jaw, which minimises stress concentrations and ensures uniform stress distribution across the entire specimen surface. The loading arrangement is prominently depicted in Fig. 3(a), while a portion of the loading-unloading sequence is illustrated in Fig. 3(b). Each cycle of the investigation consisted of applying a load up to a specified percentage of the specimen's ultimate capacity, followed by unloading to an unstressed condition, and subsequently performing ultrasonic evaluation. Such step loading is designed to track the evolution of nonlinearity with accumulating damage. A uniform loading and unloading rate of 50 ± 5 kN/min was maintained throughout. This sequence was repeated until 90 % of the specimen's ultimate capacity was reached, as shown in Fig. 3(b). Therefore, there were nine loading-unloading conditions in addition to the no-load condition. Following each unloading step, the ultrasonic wave evaluation was conducted.

### 4.3. Excitation and measurement of ultrasonic waves

The experimental setup comprises a National Instruments (NI) PXIe-1073 system connected to a computer. This system includes a NI PXI-5412 arbitrary waveform generator. The generated signal is fed into a Ciprian Amplifier HVA-800-A, where the voltage is amplified by a factor of 400. In this study, the input voltage from the NI system ranges from 0.1 V to 0.4 V, with increments of 0.1 V. It is then amplified to 40 V, 80 V, 120 V, and 160 V, respectively, to ensure sufficient energy for the bulk wave to propagate through the specimen. Prior to each experiment, voltage amplification calibration was performed.

Two ULTRAN direct contact transducers (GC200-D13 and GC350-D13), with a probe diameter of 12.7 mm, were used to transmit and receive the ultrasonic bulk waves through the specimen along its 40 mm dimension. The same experimental setup was employed for both linear and nonlinear ultrasonic techniques. The signal was measured and recorded using a NI PXI-5105 digitiser within the NI system. A sampling frequency of 10 MHz was employed to achieve finite resolution in arrival time differences, while the vertical range was optimised to 200 mV (peak-to-peak) to ensure high-resolution signal capture. The signals were averaged 300 times to enhance the signal-to-noise ratio, resulting in a stable waveform in the received time domain signal. The procedure is illustrated in the flow diagram shown in Fig. 4.

In ultrasonic testing using contact-based transducers, it is important to maintain consistent contact during testing after each conditioning for accurate comparison of the results. To ensure reliable and uniform contact throughout the experiment, a bespoke transducer holder was developed as shown in Fig. 5. This apparatus consists of a pair of 5 mm-thick plates, springs, 6 mm diameter threaded rods, and a 3D-printed polypropylene bracket (Fig. 5(a) and (b)). The brackets secure the transducers against the specimen, while the springs provide a consistent contact force (Fig. 5(c)). This setup enables precise measurements and significantly improves the overall quality of the ultrasonic data.

### 4.4. Signal processing for determining ultrasonic parameters

The raw signal data recorded on the computer were processed to extract linear and nonlinear ultrasonic parameters following the procedures outlined in the subsequent subsections.

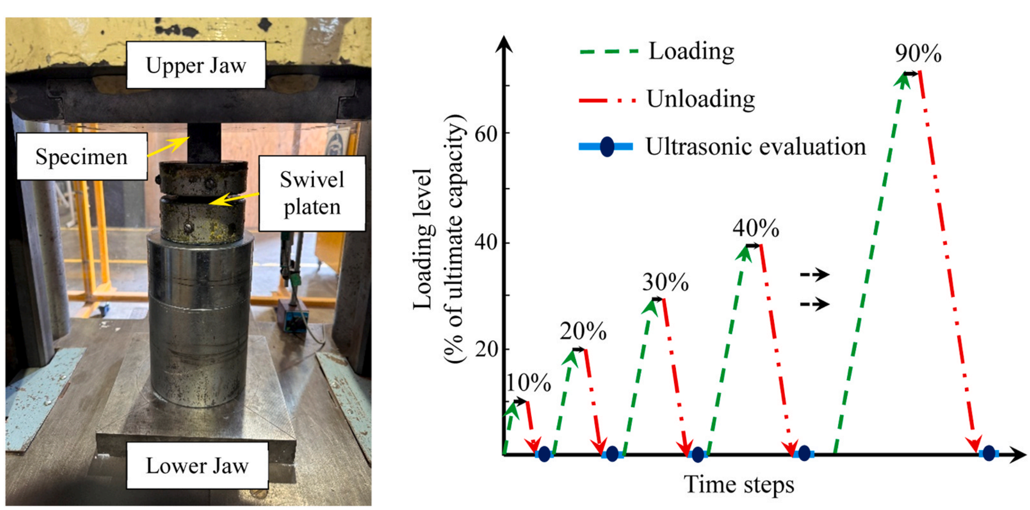
#### 4.4.1. Estimation of time-of-arrival

The time-of-arrival (ToA) is a fundamental linear ultrasonic parameter that quantifies the time delay between the initial actuation and the subsequent reception of the wavelet. A 4-cycle Hann-windowed tone burst signal was used as the excitation pulse, with the time lag up to the centre of the wave packet denoted as $t_a$. After a series of preliminary tests, a frequency of 200 kHz was selected to ensure distinguishable wave packets in the time domain—minimising overlap and reducing the influence of boundary reflections on the incident wave. The incident wave and boundary reflections observed in the received signals are illustrated in Fig. 6(a). The amplitude of the boundary reflection components decreased significantly, indicating substantial material attenuation and energy loss within the medium. After completing each loading level, at least five measurements were taken by detaching and reattaching the transducers on the surface of the specimen to account for variability in coupling. The ToA of the bulk wave travelling from one surface to the opposite surface of the specimen was determined using the signal envelope. The Hilbert Transform was applied to extract the envelope of the received signal, facilitating accurate arrival time estimation of the ultrasonic wave [30,64].

$$ToA = \left(t_1 + \frac{t_2 - t_1}{2}\right) - t_a \tag{8}$$

$$Wave\ velocity = \frac{Wave\ propagation\ distance}{ToA} \tag{9}$$

Moreover, Fig. 6(a) illustrates that boundary reflections may




(a) (b)

**Fig. 3.** Loading arrangement for (a) specimen under compressive loading in UTM, (b) loading-unloading scheme.

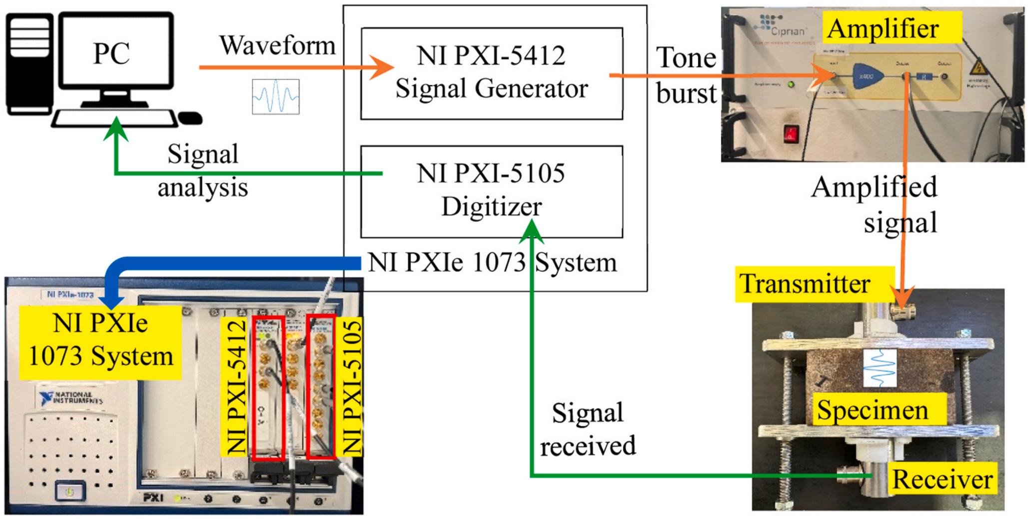



**Fig. 4.** Experimental setup for ultrasonic wave generation and measurement.

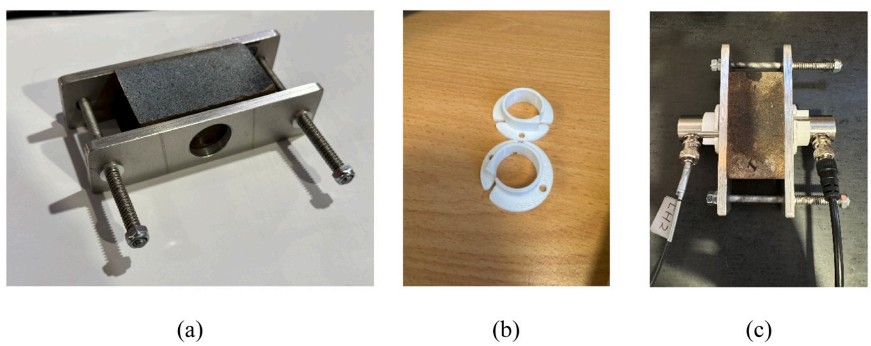



**Fig. 5.** Test specimen set-up (a) Specimens placed inside the mechanical clamping apparatus, (b) 3D-printed circular brackets, and (c) specimen arrangement along with actuator and receiver transducers.

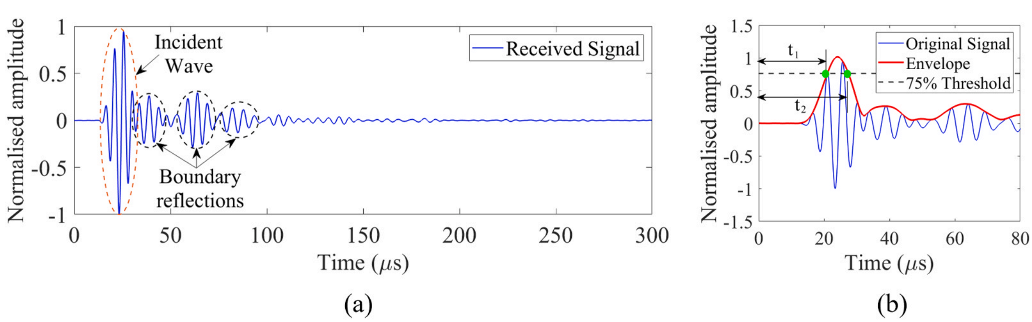



**Fig. 6.** (a) Received signal and (b) ToA calculation from signal envelope.

interfere with the incident wave, potentially affecting the peak of the wave packet. To mitigate this effect, a threshold line at 75 % of the maximum amplitude was introduced, as shown in Fig. 6(b). The intersection points between the envelope of the first-arriving wave packet of the received signal and the threshold line were identified and labelled as $t_1$ and $t_2$, respectively. Subsequently, the arrival time of the incident wave was determined using Eq. (8). By utilising the ToA and the distance between the transducers, the velocity was determined using the fundamental distance-velocity formula presented in Eq. (9).

*4.4.2. Measurement of second harmonic generation*

Higher frequency waves have shorter wavelengths, which enhances defect detection. However, they also have significant energy attenuation in concrete-like solid heterogeneous materials due to the presence of micropores and voids, by which scattering and absorption effects can reduce the amplitude of higher harmonics [33,65,66]. Therefore, careful ultrasonic feature selection is essential to enhance the accuracy and reliability of damage detection. In this study, after conducting multiple trials, a 100 kHz excitation frequency was identified as optimal for nonlinear ultrasonics. It also enables the generation of well-spaced frequency peaks. Similar to the approach used in linear ultrasonic studies, a 4-cycle Hann-windowed tone burst signal at 100 kHz was utilised for excitation. Fig. 7(a) presents a typical time-domain response of the received signal at 0 % loading. The waveform contains several consecutive reflected components that cannot be separated from the incident wave. The smaller cross-section of the specimen resulted in shorter

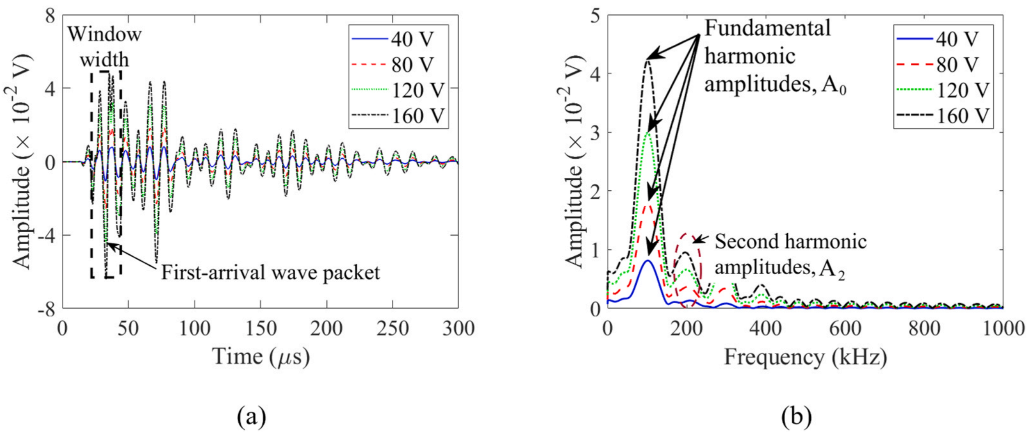



**Fig. 7.** (a) Time domain of received signal, and (b) FFT of received signal showing fundamental and second harmonic components.

propagation paths, leading to multiple reflections within the specimen. The incident wave may also generate harmonics, even when interacting with boundary reflections, as theoretically demonstrated by Chen et al. [67] and further described in [68].

In order to examine the second harmonic generation, the received signal was transformed into the frequency domain using a fast Fourier transform (FFT) [69]. However, isolating the first arrival wave packet in the time domain is difficult because multiple boundary reflections interfere with the signal. This interference can be due to the use of small-sized specimens and/or lower excitation frequency. A rectangular window was applied to the signal extracting data from $t_{start} = 22.5$ μs to $t_{end} = 45.5$ μs as shown in Fig. 7(a), before performing the FFT. This range was deliberately chosen to capture the first-arriving wave packet while excluding the initial few cycles, thereby minimising the possible disturbances from transient and ringing effects [70,71]. The extracted signal was zero-padded to achieve a 20,000-sample length, thus enhancing frequency resolution. Accordingly, the same window length and position were consistently maintained throughout the data analysis of each loading step to prevent any bias in result comparisons.

Fig. 7(b) presents the frequency domain of the received signal at 0 % loading level for four different voltage inputs. Both the fundamental harmonic amplitude ($A_0$) and the second harmonic amplitude ($A_2$) increase with voltage. This allows for the establishment of a linear relationship between $A_2$ and $A_0^2$ through the data obtained from a series of measurement points. Second harmonic generation is observed in the frequency domain, with the harmonic amplitude notably large despite the absence of applied load during the experiment. This phenomenon may be attributed to the presence of pores, air voids, and pre-existing cracks within the specimens.

Additionally, the time domain signal was also analysed using the time-frequency spectrum, which does not necessitate selecting a fixed window position. The time-frequency spectrum was derived using a short-time Fourier transform (STFT) on the time domain. STFT provides a consistent distribution of the windowing function and segments. It employs a short-duration time window and applies the Fourier transform to the windowing function that sweeps across the entire time domain of the received signal [72]. In this study, the window interval is carefully selected by applying a Hann window to ensure sufficient information regarding the fundamental and second harmonic frequencies of the measured signal is retained [30]. Through this process, the second harmonics can be extracted and used to compare with the FFT results. The spectra in Fig. 8(a) clearly illustrate harmonic generation, while Fig. 8(b) presents the spectral amplitude at the fundamental and second harmonic frequencies. This analysis effectively reveals the presence of harmonic generation and its spatial occurrence. The peak spectral amplitude corresponding to the first arriving wave packet was considered as the fundamental harmonic amplitude ($A_0$), while the peak at its

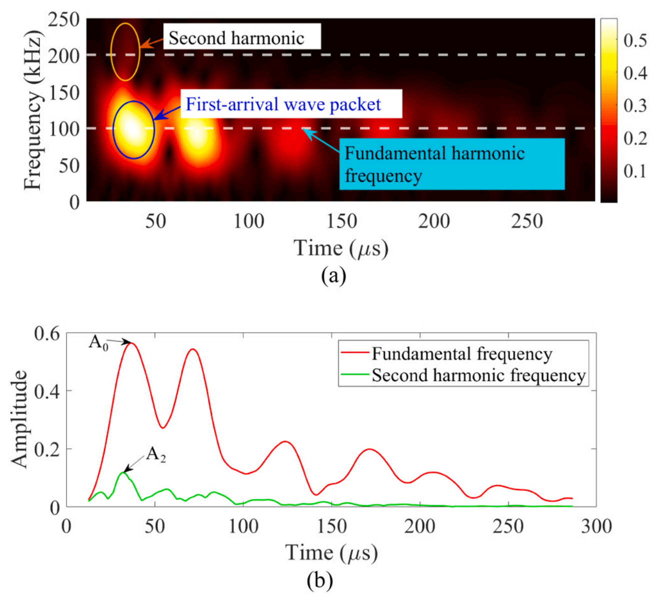



**Fig. 8.** (a) Time-frequency spectrum, and (b) spectral amplitude at fundamental ($f_0$) and second harmonic frequencies ($2 f_0$).

second harmonic frequency was taken as $A_2$.

## 5. Results and discussion

### 5.1. Variation in wave velocity and amplitude at different loading levels

The wave velocity behaviour at different loading levels is a critical factor to explore, as the stress-strain response of RRC does not follow a perfect linear relationship throughout the entire curve [15,20]. The ToA of the incident wave was determined after each loading-unloading cycle and converted to wave velocity using Eq. (9), which is presented in Fig. 9. The average bulk wave velocity measured through the unloaded specimen was 2870.8 m/s. Upon applying 50 % of the ultimate load, a relatively consistent decreasing trend is observed in the averaged wave velocity. The initial unexpected pattern observed in the plot of wave velocity against percentage loading level is likely due to the applied load being insufficient to generate significant cracks, resulting in a minimal percentage change [73]. Moreover, the changes in wave velocity are comparable to the error bars shown in the figure. Although the graph shows several fluctuations, the overall trend indicates minimal change in wave velocity at lower loading levels, followed by a relatively larger reduction after 50 % loading. The linear trend exhibits a steady decline,

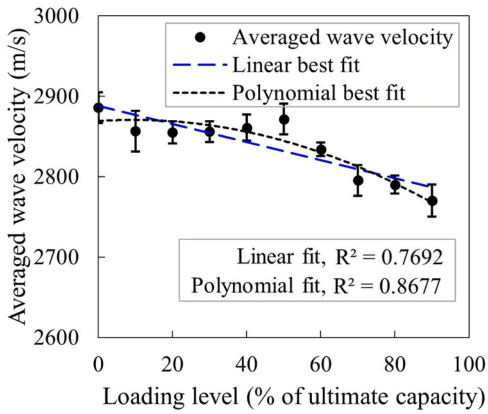



**Fig. 9.** Trend of average wave velocity with different levels of applied compression loading.

whereas the polynomial trend begins with a constant phase before transitioning into a downward trend. The polynomial trend appears to capture the situation better.

Amplitude is another important linear ultrasonic parameter for assessing damage. Fig. 10 presents the measured amplitude at various loading levels across four amplified voltage inputs: 160 V, 120 V, 80 V, and 40 V. Overall, the trends indicate a decrease in signal amplitude as the load increases. The changes in amplitude under maximum loading conditions, compared to the non-load state, are 11.2 %, 19.9 %, 3.3 %, and 10.3 % for the 160 V, 120 V, 80 V, and 40 V inputs, respectively. Among these, the 120 V input shows significant sensitivity and consistency as the load increases, making it the preferred choice for further comparison. The initial increase in amplitude observed in the 160 V case may be attributed to discrepancies in contact between the transducer and the specimen.

In addition to velocity and amplitude, linear ultrasonic parameters related to frequency, such as frequency shifts, were also investigated by converting the time-domain data to the frequency domain. The central frequency exhibited no significant shift between the intact and 90 % load conditions and was not found to be as sensitive as the other linear ultrasonic parameters discussed above.

### 5.2. Second harmonic generation at different loading levels

The amplitudes $A_0$ and $A_2$ were computed from the time domain of the received signal by using FFT analysis as described in Section 4.4.2. Fig. 11 illustrates the relationships between the second harmonic amplitude ($A_2$) and the square of the fundamental harmonic amplitude ($A_0^2$) at four different voltage inputs and at varying load levels. The $A_2$ versus $A_0^2$ relationships exhibit mostly linear increasing trends across all

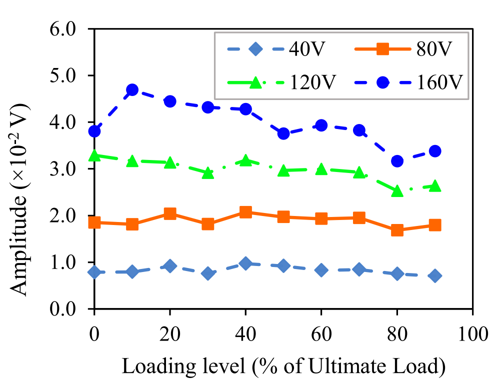



**Fig. 10.** Measured amplitude at different loading levels with different voltages of excitation signals.

loading levels, which aligns with previous studies [44,74]. The coefficient of determination ($R^2$) ranges from 0.93 to 0.99, indicating a strong correlation between $A_2$ and $A_0^2$. The slopes of these linear fits serve as relative measures of the nonlinearity parameter (β) for each loading level [75]. Furthermore, in this study, $A_2$ increases linearly with $A_0^2$ as voltage increases, following a trend similar to previous studies and confirming the growth of second harmonic generation [37,45,76].

The relative nonlinearity parameters (β') for each loading level were determined by analysing the slope of the linear relationships between $A_2$ and $A_0^2$. The presence of voids, pores, and pre-existing cracks results in non-zero values even under unloaded conditions. Additionally, it can be seen from the figures that the trend lines are slightly offset from the origin, which can be attributed to the presence of system nonlinearities. Thus, normalisation assists in eliminating these initial ambiguities, providing a more effective means of interpreting the impact of loading. Subsequently, the data points were normalised relative to the 0 % loading level. Here, as evident from Fig. 12, the relative nonlinearity parameter serves as a reliable indicator for distinguishing damage progression across various loading levels. A slight decreasing trend is observed up to a 20 % loading level, indicating that the material remains within its elastic range under this initial minimal load. However, as the compression load increases, voids and pre-existing cracks begin to close, leading to a small reduction in the nonlinearity parameter. This is a behaviour commonly observed in conventional concrete [37,76].

After applying a load of 20 % of the ultimate capacity, microcracks may begin to form and progressively grow, which potentially leads to a continuous increase in the relative nonlinearity parameter. At moderate loading levels, up to 40 %, the data points remain clustered around the fitted curve. However, beyond this point, they start to deviate. This deviation may be attributed to the rapid formation of microcracks, which eventually evolve into macrocracks, causing increased wave scattering. It may also result from the repeated attachment and detachment of the transducers. Nevertheless, it has an overall increasing trend. An upper and lower boundary is established using the average deviation to assess the variability of the data. These bounds, represented as dashed lines, enclose most data points, highlighting the inherent variability within the dataset. The mean absolute deviation from the trend line is 11 %, indicating the average extent to which the data points deviate from the expected trend.

Likewise, STFT was applied further to analyse the measured signals in the time-frequency domain. This technique allows observation of how the frequency components in the signal change over time. The spectral amplitude data ($A_0$ and $A_2$) were extracted from the time-frequency spectrum as described. Similar to the previous representation, the $A_2$ versus $A_0^2$ relationships closely resemble those from the FFT analysis, but with better correlation. However, this approach is more convenient as it eliminates the need to fix the window location. The relative nonlinearity parameters were obtained from the slope of the $A_2$ versus $A_0^2$ relationships and plotted in Fig. 13 after normalisation. The best-fit linear curve exhibits a trend similar to that observed in the FFT analysis. The upper and lower boundary limits of the scattered data points are also presented here. The mean absolute deviation from the fitted curve is 6 %, indicating a good correlation. When comparing this dataset to Fig. 12, it is evident that the scattered data points are more closely clustered around the fitted trend curve, indicating greater reliability.

### 5.3. Sensitivity and reliability assessment of ultrasonic parameters

In the previous sections, ultrasonic parameters were determined through a comprehensive analysis of wave propagation in the RRC samples under varying levels of compressive loads. It was, however, established that the absolute values of both linear and nonlinear parameters across different loading cases cannot be directly compared, as ultrasonic features, such as frequency, varied throughout the experiment. To evaluate the sensitivity of both linear and nonlinear ultrasonic

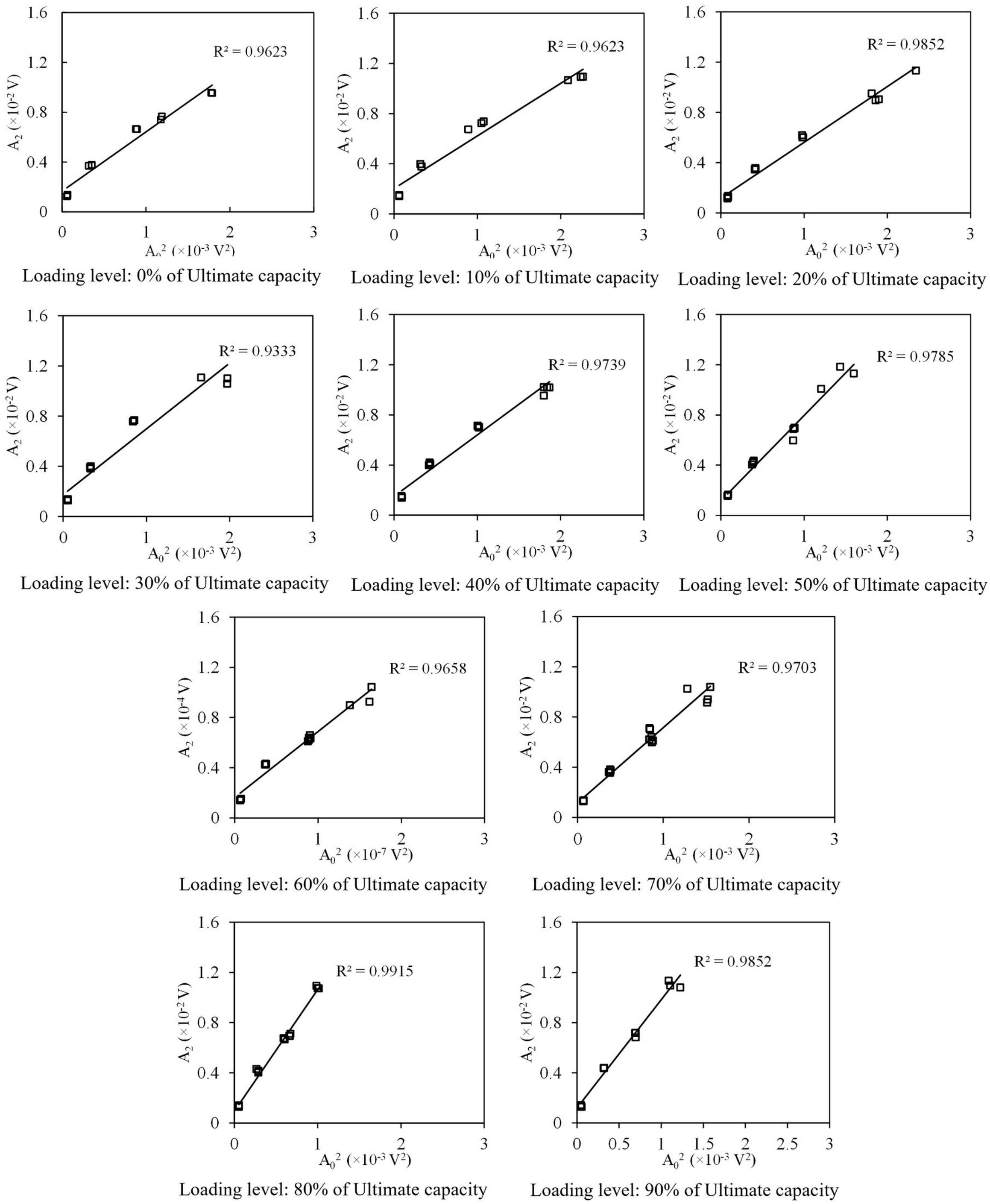



**Fig. 11.** Linear relationship between $A_2$ and $A_0^2$ with increasing voltage from FFT analysis.

parameters to progressive damage, the parameters were normalised using the results from the 0 % loading level as a baseline.

In Fig. 14, the normalised linear and non-linear parameters are plotted against increasing load levels, with a linear trend line connecting all scattered data points. The trend lines for linear parameters exhibit a distinct decreasing pattern as loading levels rise, whereas nonlinear parameters show an increasing trend, highlighting the contrasting behaviours of these two categories. At the maximum load level (90 % of ultimate capacity), wave velocity changes minimally by only 4.0 %. In contrast, amplitude undergoes a substantial reduction of up to 19.9 % compared to the baseline. This pronounced decrease in amplitude, being significantly greater than that of the wave velocity, provides a more effective indicator of damage progression.

On the other hand, the nonlinearity parameters obtained from both FFT and STFT analyses exhibit consistent trends, reinforcing the reliability of the findings. The similarity in trends across different loading levels underscores the importance of employing multiple analysis techniques to achieve a comprehensive understanding of the material response. Notably, both nonlinearity parameters show a significant change after reaching a loading level of 20 %. The maximum increase in nonlinearity parameters, derived from FFT and STFT analyses at the highest applied load, is approximately 76 % and 65 %, respectively. Unlike linear parameters, the nonlinearity parameters display a substantially distinct trend with increasing loading levels, highlighting their

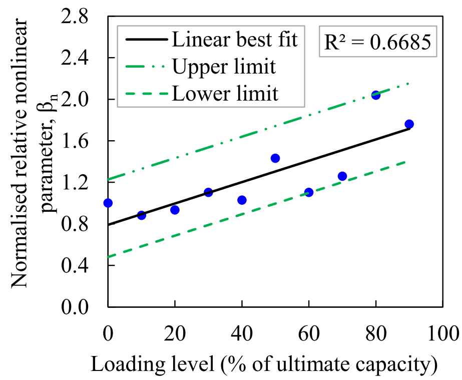



**Fig. 12.** Normalised relative nonlinearity parameter extracted from FFT analysis at varying load levels.

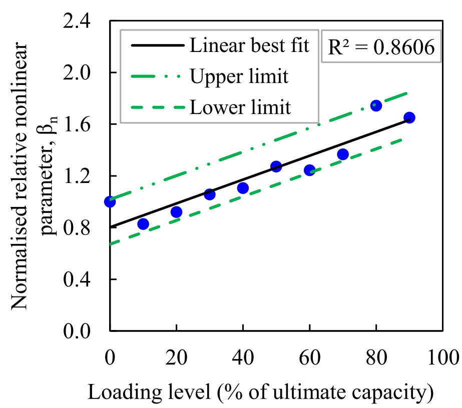



**Fig. 13.** Normalised relative nonlinearity parameter extracted by STFT at varying load levels.

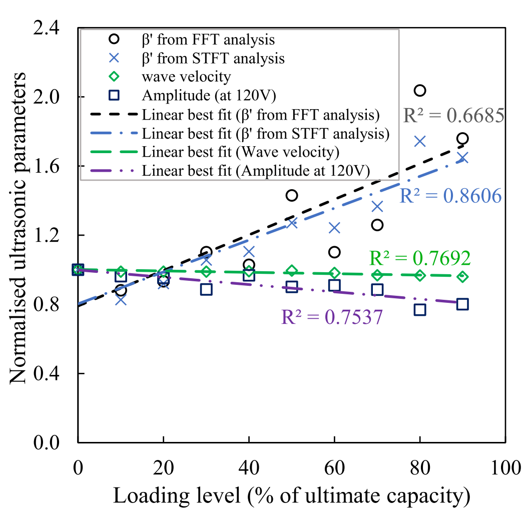



**Fig. 14.** Normalised ultrasonic parameters with respect to different levels of loading.

greater sensitivity to material damage.

To evaluate the reliability of both linear and nonlinear parameters, the coefficient of determination ($R^2$) for each trend line was obtained from Fig. 14. While the nonlinearity parameters derived from FFT analysis exhibit strong sensitivity, the linear parameters show less variation in the data. The choice of window position may influence the variability observed during the FFT analysis, which has been addressed by adopting the STFT. The $R^2$ value was found to be as high as 0.86 for the nonlinearity parameters obtained from STFT analysis. These findings indicate that the nonlinearity parameter is more reliable in assessing material behaviour under varying loading conditions.

## 6. Conclusions

This paper presented a comprehensive study on assessing damage progression by applying different levels of compressive load in a new material derived from regolith simulant for potential use in lunar construction. Nine distinct load levels were applied to the specimens. Simultaneously, linear and nonlinear ultrasonic analyses were conducted before and after each loading step to evaluate the effectiveness of these parameters in damage assessment. Among the linear parameters examined in this study, it was found that amplitude was more sensitive to progressive damage than the wave velocity. While lower voltage levels did not cause significant changes in amplitude, a 120 V power input resulted in approximately a 19.9 % reduction. In addition, the nonlinearity parameter derived from second harmonic generation effectively distinguishes the effects induced by different loading levels, demonstrating the technique's sensitivity in capturing material responses. Based on sensitivity and reliability analysis, the nonlinear ultrasonic parameter proves to be significantly more effective than the linear parameter. Moreover, the nonlinearity parameter exhibits a consistent increase after applying 20 % of the ultimate load.

Two different techniques were used to extract the nonlinearity parameter from the received signal. While FFT analysis requires selecting a specific window location, STFT provides a moving window approach. Although the nonlinearity parameters obtained from STFT yield more reliable results, both analyses demonstrate comparable sensitivity. Thus, the results from both methods reinforce and validate each other. In summary, the findings of this study highlight that second harmonic generation techniques serve as an effective method for monitoring damage progression in the RRC materials for lunar construction. However, the behaviour of ultrasonic techniques should differ under lunar environmental conditions. Therefore, to address this, future work should investigate the behaviour of ultrasonic waves in RRC under simulated lunar environmental conditions, including vacuum as well as high and low temperatures.

## CRediT authorship contribution statement

**Smith Scott:** Writing – review & editing, Supervision. **Tafsir Tafsirojjaman:** Writing – review & editing, Supervision. **Ching Tai Ng:** Writing – review & editing, Supervision, Conceptualisation. **Md Maruf Molla:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualisation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

A strategic off-earth construction PhD scholarship awarded by Adelaide University to the first author. The support is gratefully acknowledged. Additionally, the authors would like to thank Mr Mohammad Altaf Hossain for his assistance with specimen preparation for this study.

## Data Availability

Data will be made available on request.

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